Free multi-domain diffusion models and the thermal history of Win-06-01b goethite from the water-saturated step heat HT-Win-06-01b

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Abstract

We fit the ^{3}He release of the water-saturated step-heating experiment on proton-irradiated Win-06-01b goethite with multi-domain diffusion (MDD) models of $N = 1$–10 spherical domains whose ^{3}He fractions and sizes are all free, following Farley and Flowers (2012). Each domain structure is then used to fit a burial-and-exhumation thermal history to the bulk-normalized ^{4}He/^{3}He spectrum and to predict the (U–Th)/He age. The ^{3}He yields are best described (ΔBIC) by $N = 3$; beyond $N = 4$ added domains duplicate existing ones and the fit does not improve. Every $N \geq 2$ model places ≈97% of the ^{3}He in one domain slightly larger than the Stage 1 reference domain ($r/r_0 \approx 1.03$–1.12) and 2–3% in small, fast-degassing domains. The best burial paths peak at 60–65 °C (joint 1σ ranges 57–67 °C) between ∼60 and 100 Ma, consistent with regional apatite fission-track constraints. However, the domain structures preferred by the ^{3}He data ($N \geq 3$) fit the ^{4}He/^{3}He spectrum markedly worse than the two-domain structure, which exposes a tension between the ^{3}He and ^{4}He data that a single-activation-energy, diffusion-only model does not resolve.

1 Motivation and data

Win-06-01b is dense colloform goethite collected as float in the Flinders Ranges, South Australia, with a (U–Th)/He age of 130–180 Ma (Waltenberg, 2012). Apatite fission-track studies in the region record cooling from roughly 50–70 °C after the mid-Cretaceous (Gibson and Stüwe, 2000; Mitchell et al., 1998), and Triassic intramontane basins less than 1.5 km thick indicate that parts of the ranges were near the surface in the Early Mesozoic (Mitchell et al., 1998). A proton-irradiated aliquot was step-heated in a sealed silica ampoule under water-saturated conditions (HT-Win-06-01b; 34 steps, 170–250 °C), releasing 14.8% of its ^{3}He before final fusion. The Stage 1 regression of that experiment defines the reference Arrhenius line used here ($E_a = 157.1$ kJ mol^{-1}, $\ln(D_0/a_0^2) = 16.16$ with D_0/a_0^2 in s^{-1}).

Earlier versions of this analysis split the sample into a high-retentivity domain pinned to the Stage 1 line and a low-retentivity assemblage holding at most 7.5% of the ^{3}He. Here that distinction is dropped: all domains are equivalent, and each has a free ^{3}He fraction and size.

2 Methods

2.1 Domain model

The sample is represented by N non-interacting spherical domains that share the Stage 1 activation energy and frequency factor. Domain j holds a fraction f_j of the uniformly distributed, proton-

induced ^{3}He ($\sum_j f_j = 1$) and has radius r_j relative to the reference radius r_0 implied by the Stage 1 regression, so that

$$\left(\frac{D}{a^2}\right)_j = \left(\frac{r_j}{r_0}\right)^{-2} \frac{D_0}{a_0^2} \exp\left(-\frac{E_a}{RT}\right). \tag{1}$$

A model with N domains has $k = 2N-1$ free parameters. Sizes are bounded by $10^{-6} \le r_j/r_0 \le 10^3$.

2.2 Laboratory release of ^{3}He

For a heating schedule of steps i with temperatures T_i and durations Δt_i, the reference domain accumulates the non-dimensional time

$$\tau_{0,i} = \sum_{m \le i} \frac{D_0}{a_0^2} \exp\left(-\frac{E_a}{RT_m}\right) \Delta t_m, \tag{2}$$

and domain j reaches $\tau_{0,i}(r_j/r_0)^{-2}$. An initially uniform sphere releases the cumulative fraction (Crank, 1979)

$$F(\tau) = 1 - \frac{6}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} \exp\left(-k^2\pi^2\tau\right), \tag{3}$$

so the model cumulative and fractional ^{3}He yields of the sample are

$$\Sigma F_i = \sum_{j=1}^{N} f_j \, F\left(\tau_{0,i}\,(r_j/r_0)^{-2}\right), \qquad \Delta F_i = \Sigma F_i - \Sigma F_{i-1}. \tag{4}$$

2.3 Fitting the domain structure to ^{3}He

Following Farley and Flowers (2012, supplement A2), the fractional and cumulative yields are weighted equally, and $\{f_j, r_j\}$ minimize

$$S = \sum_{i=1}^{34} \left(\Delta F_i^{\mathrm{mod}} - \Delta F_i^{\mathrm{obs}}\right)^2 + \sum_{i=1}^{34} \left(\Sigma F_i^{\mathrm{mod}} - \Sigma F_i^{\mathrm{obs}}\right)^2, \tag{5}$$

with yields expressed as fractions of the total ^{3}He, including the fusion step. Models are built up one domain at a time: the N-domain search starts from the $(N-1)$-domain solution plus one new domain trialled at 15 sizes and two fractions, together with ten random starts. The fit therefore cannot get worse as N increases. To compare models of different complexity, we use the Bayesian information criterion for Gaussian residuals of unknown variance,

$$\mathrm{BIC} = n \ln(S/n) + k \ln n, \qquad n = 68. \tag{6}$$

The Arrhenius and delta plots are not fitted. They follow from applying the single-sphere inversion of Fechtig and Kalbitzer (1966) to the model cumulative release. Their misfit is reported as a reduced χ^2 against the bounding-method uncertainties in $\ln(D/a^2)$.

2.4 Radiogenic ^{4}He in nature

Each domain produces ^{4}He uniformly at rate P and loses it by diffusion with a zero-concentration boundary,

$$\frac{\partial C}{\partial t} = \frac{D}{a^2} \frac{1}{x^2} \frac{\partial}{\partial x}\left(x^2 \frac{\partial C}{\partial x}\right) + P, \qquad C(1,t) = 0, \quad C(x, t_{\mathrm{form}}) = 0, \tag{7}$$

where x is radial position normalized to the domain radius and $t_{\rm form}$ is the crystallization age. With $C = \sum_k A_k(t)\sin(k\pi x)/x$ (Carslaw and Jaeger, 1959), each mode evolves as

$$\frac{\mathrm{d}A_k}{\mathrm{d}t} = -k^2\pi^2\frac{D}{a^2}A_k + c_k P, \qquad c_k = \frac{2(-1)^{k+1}}{k\pi}, \tag{8}$$

which over an interval Δt (2 Myr) at constant temperature has the exact solution

$$A_k \leftarrow A_k e^{-\lambda_k} + c_k P\,\Delta t\,\frac{1-e^{-\lambda_k}}{\lambda_k}, \qquad \lambda_k = k^2\pi^2\frac{D}{a^2}\Delta t. \tag{9}$$

We use 8000 modes. The ^{4}He content per unit volume of domain j is $N_j = 3\sum_k A_{k,j}(-1)^{k+1}/(k\pi)$. Taking U and Th to be distributed like the ^{3}He and neglecting parent decay (∼3% over 200 Myr), the bulk He age is

$$t_{\rm He} = \frac{1}{P}\sum_j f_j\,N_j. \tag{10}$$

2.5 Laboratory degassing of the natural profiles

In the laboratory each mode decays as $\exp(-k^2\pi^2\tau)$. The reference times $\tau_{0,i}$ used for the ^{4}He prediction are solved from the *measured* cumulative ^{3}He by inverting Eq. 4. The predicted spectrum is therefore independent of the heating schedule and of step-to-step kinetic scatter:

$$\frac{R_{{\rm step},i}}{R_{\rm bulk}} = \frac{\Delta N_{4,i}/N_{4,\rm tot}}{\Delta F_i^{\rm obs}}, \qquad \Delta N_{4,i} = \sum_j f_j\left[N_j(\tau_{0,i-1}) - N_j(\tau_{0,i})\right], \tag{11}$$

where $N_j(\tau)$ is the ^{4}He remaining in domain j after laboratory time $\tau(r_j/r_0)^{-2}$ and $N_{4,\rm tot} = \sum_j f_j N_j(0)$ includes the gas released at fusion, consistent with the measured bulk ratio.

2.6 Thermal histories

The thermal history is piecewise linear through $(t_{\rm form}, T_s)$, $(t_{\rm peak}, T_{\rm max})$, (50 Ma, T_s) and $(0, T_s)$ with $T_s = 20\,^\circ$C. The goethite forms at the surface, is buried and heated to a peak temperature, and is exhumed by the Paleogene, when Paleogene sediments in Flinders Ranges valleys show that basement was exposed (Mitchell et al., 1998). For each domain structure and each $t_{\rm form} \in \{180, 190, 200, 210\}$ Ma, $(t_{\rm peak}, T_{\rm max})$ are optimized by Nelder–Mead against

$$\chi^2 = \sum_i \left(\frac{R_i^{\rm mod} - R_i^{\rm M}}{\sigma_i}\right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2}\left(R_i^{\rm UB} - R_i^{\rm LB}\right), 0.02\right], \tag{12}$$

over the 24 steps that have lag-correction bounds and more than 0.1 pccSTP of ^{3}He. M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method. We also map χ^2 on a grid of $t_{\rm peak}$ (2.5 Myr steps) and $T_{\rm max}$ (1 °C steps) for $t_{\rm form}$ = 180, 195 and 210 Ma. The joint 1σ region is $\chi^2 \le \chi^2_{\rm min} + 3.53\max(1, \chi^2_\nu)$, and the predicted He age is contoured at the limits of the measured range.

3 Results

3.1 Domain structure from ^{3}He

Table 1 and Fig. 1 summarize the fits.

- $N = 1$. A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 23$).
- $N = 2$. About 2% of the ^{3}He goes into a small domain, which captures the high early diffusivities.
- $N = 3$. A third, intermediate domain ($r/r_0 \approx 0.03$, ∼1.6% of the ^{3}He) cuts the yield misfit by more than half and lowers the Arrhenius χ^2_ν from 3.2 to 1.6. BIC is lowest here.
- $N = 4$. Splitting the smallest domain improves S only slightly.
- $N \geq 5$. Added domains collapse onto the main domain's size, so these models are identical to $N = 4$ with redundant parameters.

In every $N \geq 2$ model, ≈97% of the ^{3}He resides in one domain with $r/r_0 = 1.03$–1.12, i.e. a diffusivity 0.80–0.94 times the Stage 1 line. The fast domains together hold 2.0–2.8% (Fig. 2). The Stage 1 line is therefore a slightly fast description of the dominant domain. It was regressed on bulk-inverted diffusivities that still carry a small contribution from the fast domains. Once the dominant domain may differ from the Stage 1 reference, the $N \geq 3$ models follow the delta plot closely, including the gradual decline between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.07 that models with a domain pinned to the Stage 1 line missed (Fig. 3). Only the prograde 240 and 250 °C steps in that interval lie above the model, by ∼1–1.7σ.

Table 1: Summary of the free multi-domain models. RMS is the yield misfit ($\times 10^{-4}$, fraction of total ^{3}He); "fast" is the ^{3}He fraction in domains with $r/r_0 < 0.1$; r_{main} is the size of the dominant domain. Thermal columns give the best burial path with a predicted He age of 130–180 Ma; the last two columns give the ranges of $T_{\max}$ (°C) and t_{peak} (Ma) within the joint 1σ region of the χ^2 maps that also satisfy the He age.

N	k	RMS	BIC	Arrh. χ^2_ν	fast (%)	r_{main}/r_0	^{4}He/^{3}He χ^2_ν	t_{form}	t_{peak}	$T_{\max}$	t_{He}	$T_{\max}$ (1σ)	t_{peak} (1σ)
1	1	84.06	-645.7	23.33	0.0	0.79	6.61	200	55.0	59.5	179	60–60	55.0–55.0
2	3	19.76	-834.2	3.24	2.1	1.03	1.33	200	97.8	65.0	178	63–67	87.5–100.0
3	5	8.50	-940.5	1.60	2.8	1.12	5.87	190	74.0	60.4	173	57–64	57.5–92.5
4	7	8.05	-939.4	1.58	2.8	1.12	5.59	190	74.4	60.4	173	57–64	60.0–95.0
5	9	8.05	-930.9	1.71	2.8	1.12	5.59	190	74.4	60.4	173	–	–
6	11	8.05	-922.5	1.85	2.8	1.12	5.59	190	74.4	60.4	173	–	–
7	13	8.05	-914.1	2.03	2.8	1.12	5.59	190	74.4	60.4	173	–	–
8	15	8.05	-905.6	2.24	2.8	1.12	5.59	190	74.4	60.4	173	–	–
9	17	8.05	-897.2	2.51	2.8	1.12	5.59	190	74.4	60.4	173	–	–
10	19	8.05	-888.7	2.84	2.8	1.12	5.59	190	74.4	60.4	173	–	–

3.2 Thermal histories from ^{4}He/^{3}He

The best age-consistent burial paths peak at 60–65 °C between ∼70 and 100 Ma for every $N \geq 2$. The joint 1σ regions span 57–67 °C and 57.5–100 Ma (Table 1; Figs. 4–6). Those times and temperatures agree with the fission-track window. The spectrum constrains peak temperature tightly and trades peak time against crystallization age; the He-age isopleths then exclude crystallization much before ∼200 Ma. The quality of the spectrum fit, however, depends strongly on the domain structure:

- $N = 1$. The single-domain structure cannot fit the spectrum ($\chi^2_\nu \geq 6$).
- $N = 2$. The spectrum is fitted well ($\chi^2_\nu = 1.3$–1.5) by a peak of ≈65 °C at 80–100 Ma.

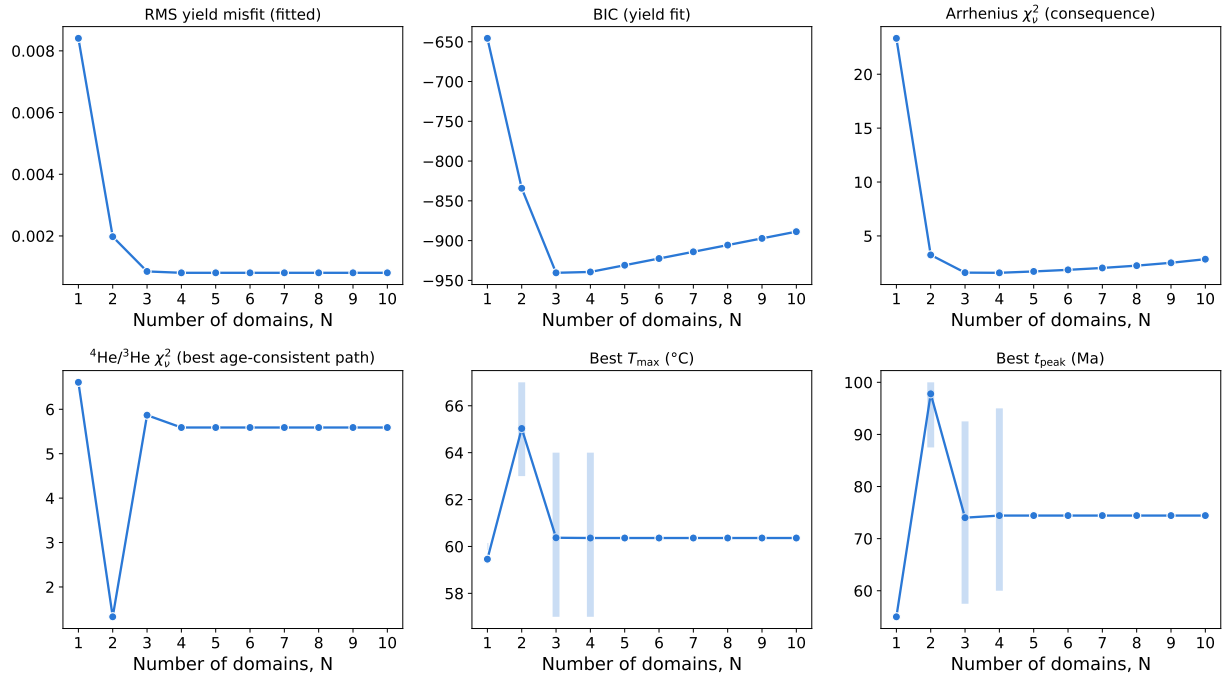


Figure 1: Fit statistics and thermal results against the number of domains. Top: RMS misfit of the fitted ^{3}He yields, BIC, and the reduced χ^2 of the resulting Arrhenius plot (not fitted). Bottom: reduced χ^2 of the best age-consistent burial path against the ^{4}He/^{3}He spectrum, and that path's peak temperature and time; shaded bars show the age-consistent ranges within the joint 1σ region of each model's χ^2 maps.

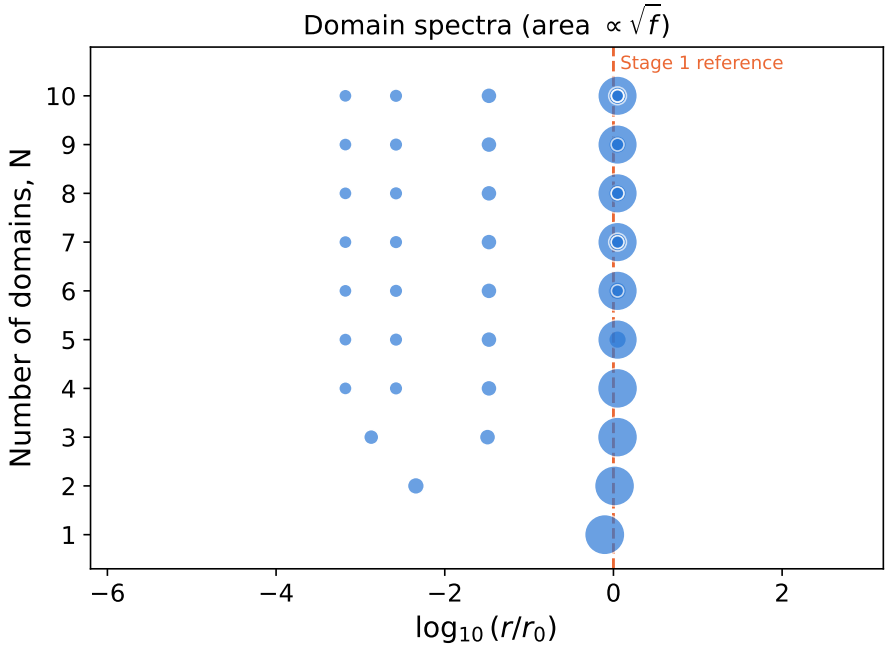


Figure 2: Domain spectra of the best-fitting models. Symbol area scales with the square root of the ^{3}He fraction; the dashed line marks the Stage 1 reference domain. From $N = 5$ onward, added domains coincide in size with the dominant domain.

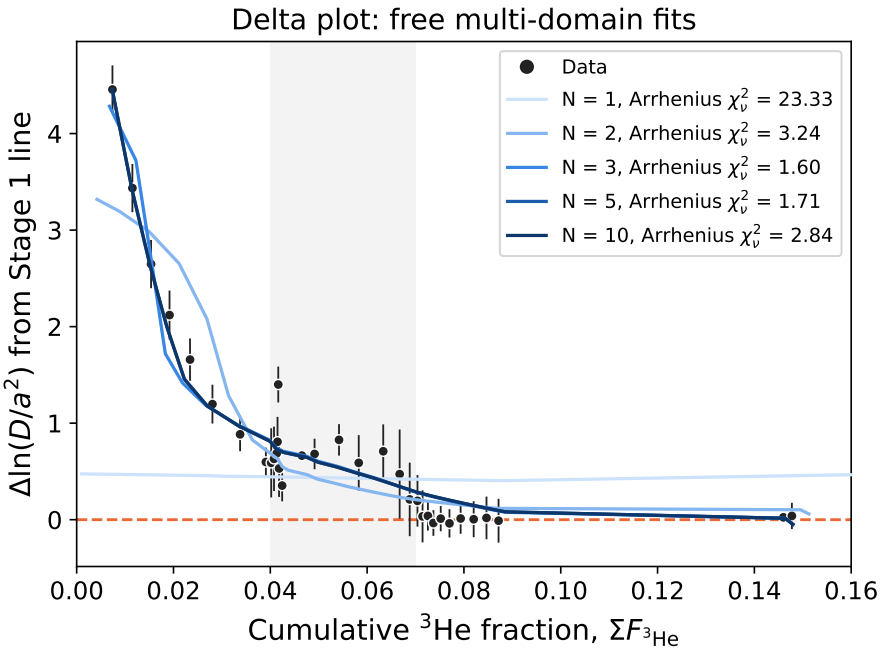


Figure 3: Delta plot of the measured diffusivities (departure from the Stage 1 line) and the curves implied by the fitted domain structures (not fitted directly). The shaded band marks $\Sigma F_{^3\mathrm{He}} = 0.04$–$0.07$, where the data show elevated diffusivities on the prograde 220–250 °C steps.

- $N = 3$ **and** $N \geq 4$**.** These are preferred by the ^{3}He data but fit the spectrum poorly ($\chi^2_\nu \approx 5.2$–6.1). Their intermediate domain ($r/r_0 \approx 0.03$) degasses at $\Sigma F_{^3\mathrm{He}} \approx 0.02$–$0.06$. With the common activation energy it would have lost most of its ^{4}He in nature, which depresses the predicted ratios exactly where the measured ratios are highest (Fig. 4).

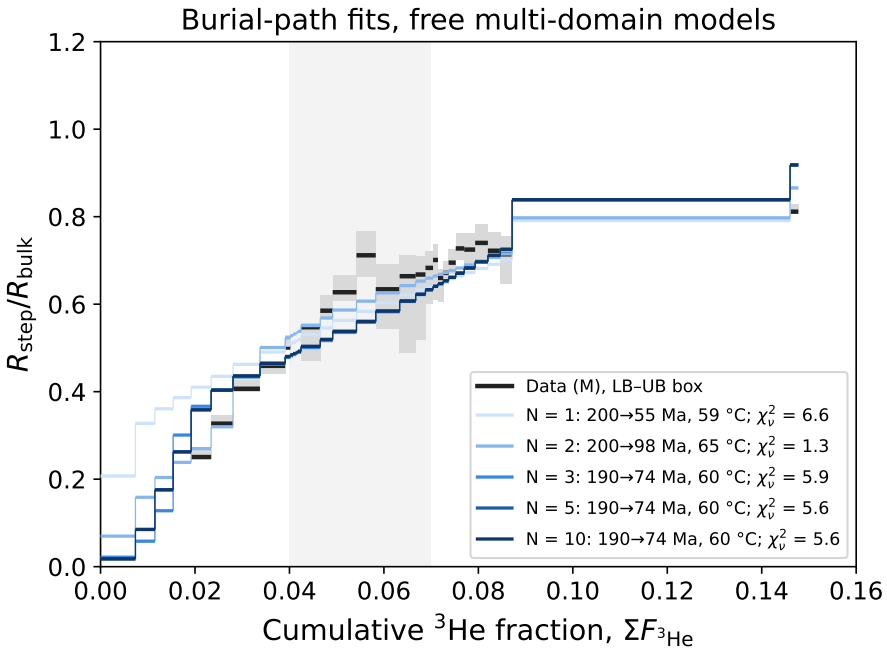


Figure 4: Bulk-normalized ^{4}He/^{3}He spectrum (black, bounding-method LB–UB boxes) and the best age-consistent burial-path predictions for selected domain structures.

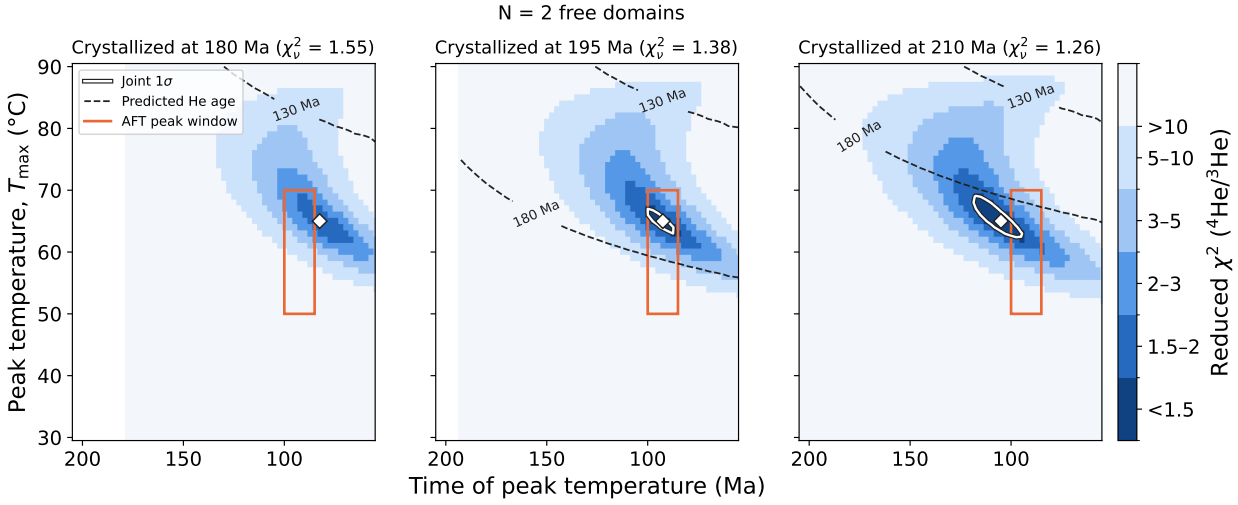


Figure 5: Reduced χ^2 of the ^{4}He/^{3}He spectrum for burial paths with the two-domain structure, as a function of the time and magnitude of peak temperature, for crystallization at 180, 195 and 210 Ma. White outlines: joint 1σ region; dashed lines: predicted bulk He age at 130 and 180 Ma; diamonds: best fit in each panel; orange boxes: apatite fission-track peak window (Gibson and Stüwe, 2000; Mitchell et al., 1998).

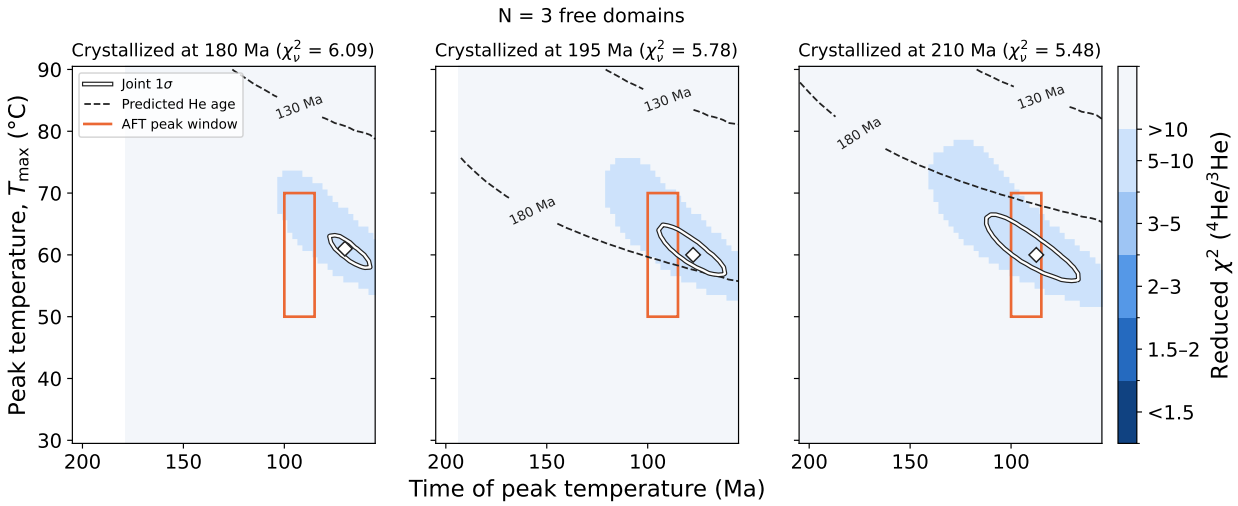


Figure 6: As Fig. 5 for the three-domain structure preferred by the ^{3}He data (lowest BIC).

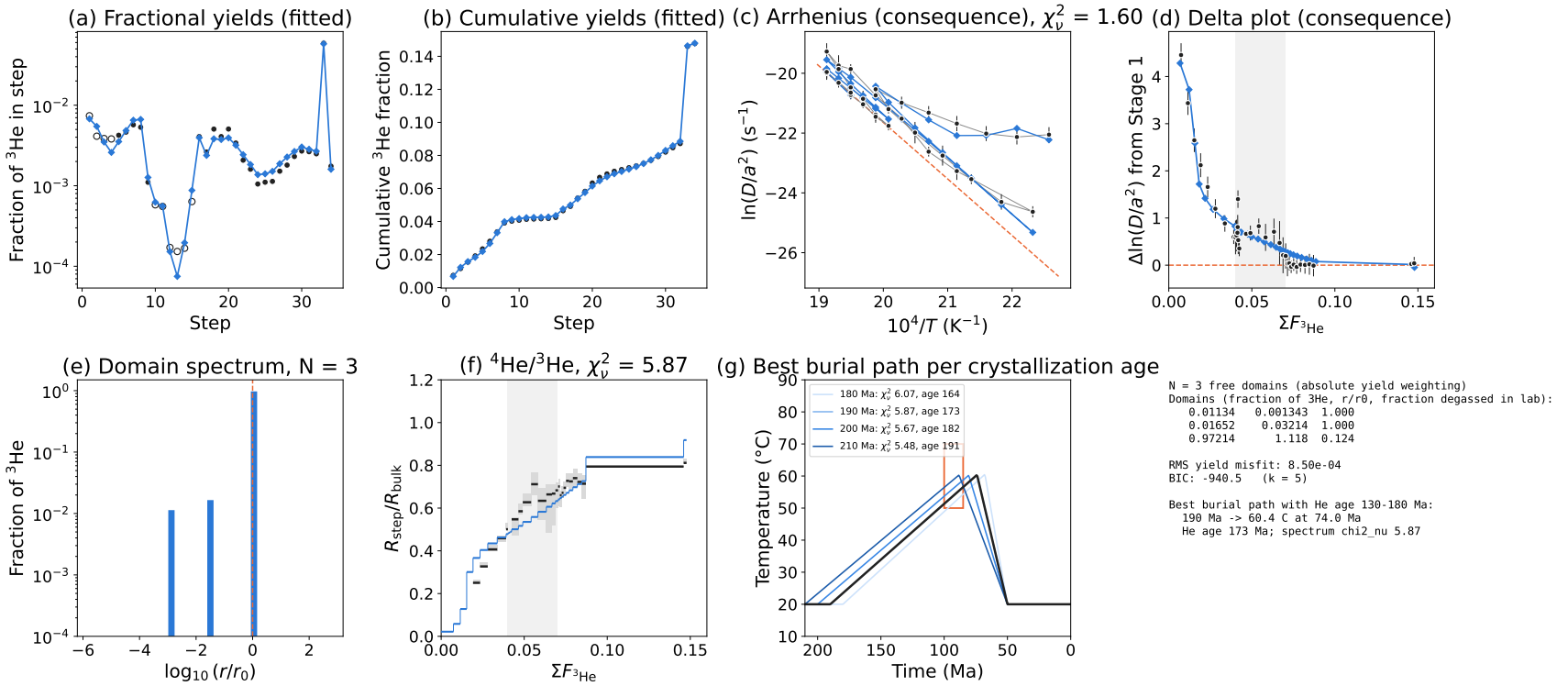


Figure 7: Diagnostic summary of the three-domain model: (a) fitted fractional and (b) cumulative ^{3}He yields; (c) Arrhenius and (d) delta plots implied by the fit; (e) domain spectrum; (f) ^{4}He/^{3}He spectrum with the best age-consistent burial path; (g) best path for each crystallization age; (h) parameter summary, including the fraction of each domain degassed during the experiment.

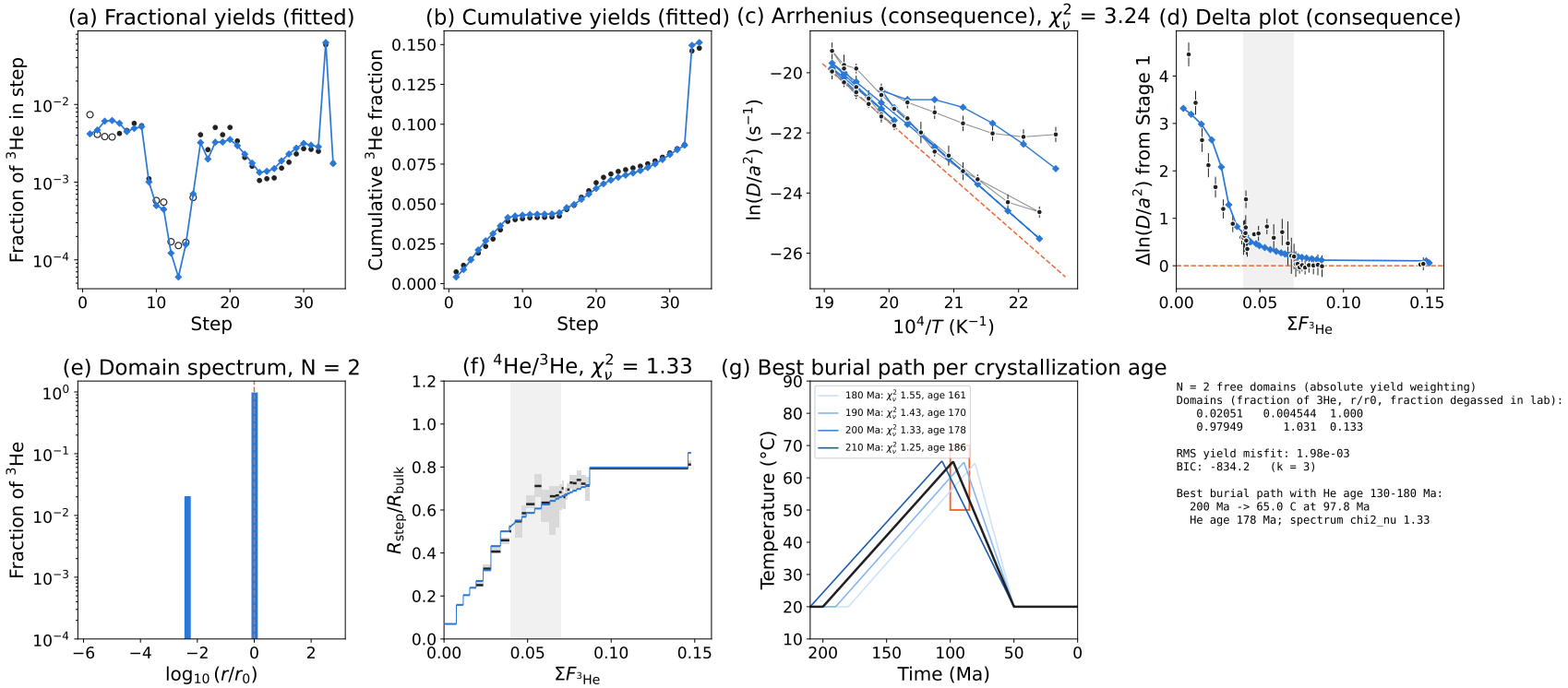


Figure 8: As Fig. 7 for the two-domain model.

4 Discussion

Robust conclusions. Across all domain structures, the dominant domain holds ≈97% of the ^{3}He and degasses slightly more slowly than the Stage 1 line. Fast domains hold only 2–3%. Every acceptable burial path peaks at 60–65 °C (joint 1σ: 57–67 °C) in the Late Cretaceous, followed by exhumation, and the He age requires crystallization at or after ∼200 Ma. This thermal history is consistent with the apatite fission-track record of 50–70 °C before cooling after ∼85 Ma (Gibson and Stüwe, 2000) and with the shallow Triassic basins and low-rank coals of the Flinders Ranges (Mitchell et al., 1998).

Tension between ^{3}He and ^{4}He. The ^{3}He yields call for an intermediate domain ($r/r_0 \approx 0.03$, ∼1.6% of the ^{3}He). With a common activation energy, that domain is too leaky in nature to supply the elevated ^{4}He/^{3}He ratios measured where it degasses. Two diffusion-only resolutions are possible:

- The intermediate domain has a higher activation energy than the Stage 1 value. It would then diffuse as fast as observed at 200–250 °C but much more slowly at near-surface temperatures, retaining more radiogenic ^{4}He. The retrograde segments of the experiment constrain the activation energy locally and could test this.
- The ^{4}He production rate differs between domains, for example through U–Th zoning across the colloform bands.

Both can be added to the present framework at the cost of one parameter each. Now that a free dominant domain largely explains the delta-plot plateau, a finite non-diffusive release during the first heating to 220–250 °C is no longer needed to explain the ^{3}He data. It remains a possible explanation for the elevated ^{4}He/^{3}He ratios in that interval.

Caveats.

- The temperature scale rests on extrapolating the Stage 1 activation energy from 200–250 °C to near-surface temperatures (± ∼4 °C near 60 °C, not propagated here).
- The exhumation time (50 Ma) and surface temperature (20 °C) are fixed.
- Domains that barely degas in the experiment are constrained only through their summed fraction.

References

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A Diagnostics and χ^2 maps for the other domain structures

Models with $N \geq 5$ are identical to $N = 4$ (added domains coincide with the dominant domain); their figures are written to `Figures/generated/mdd_free/absolute/`.

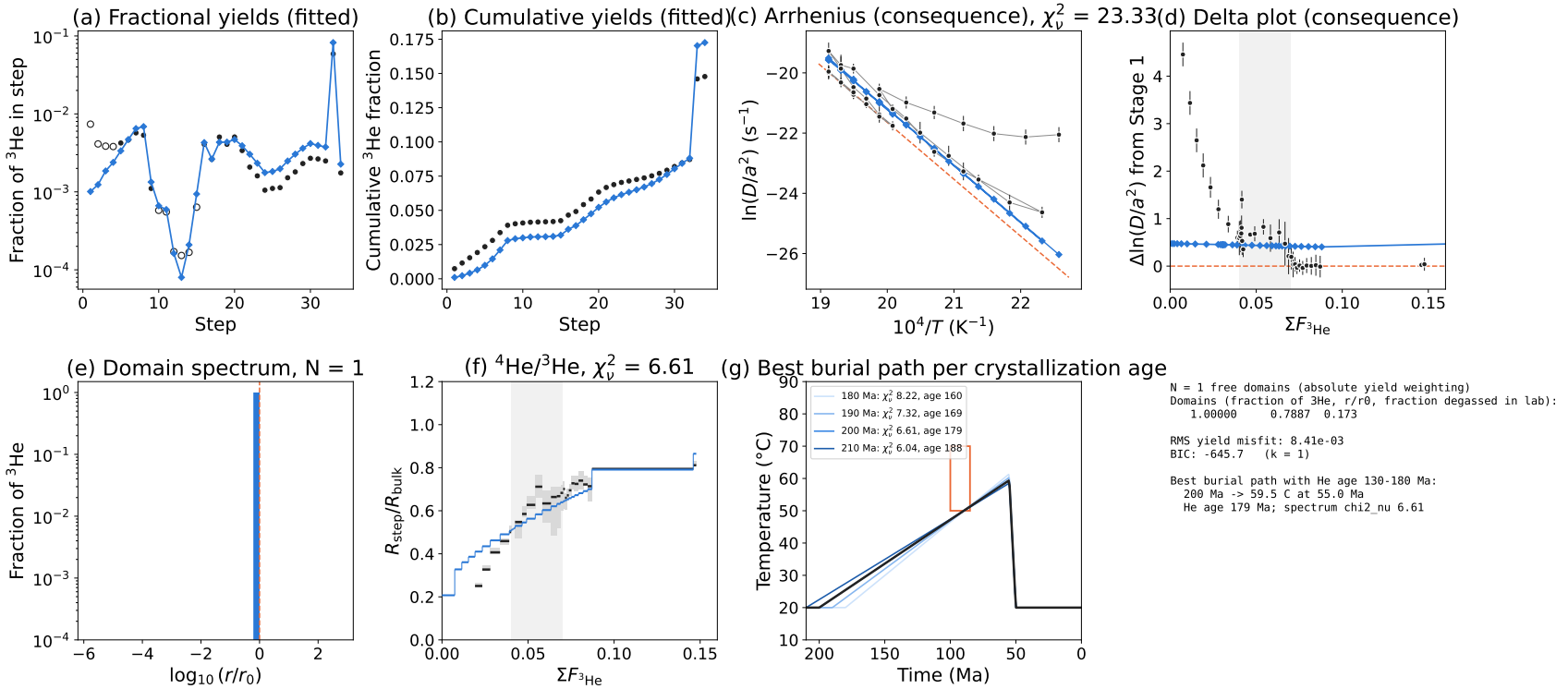


Figure 9: Diagnostic summary of the single-domain model (as Fig. 7).

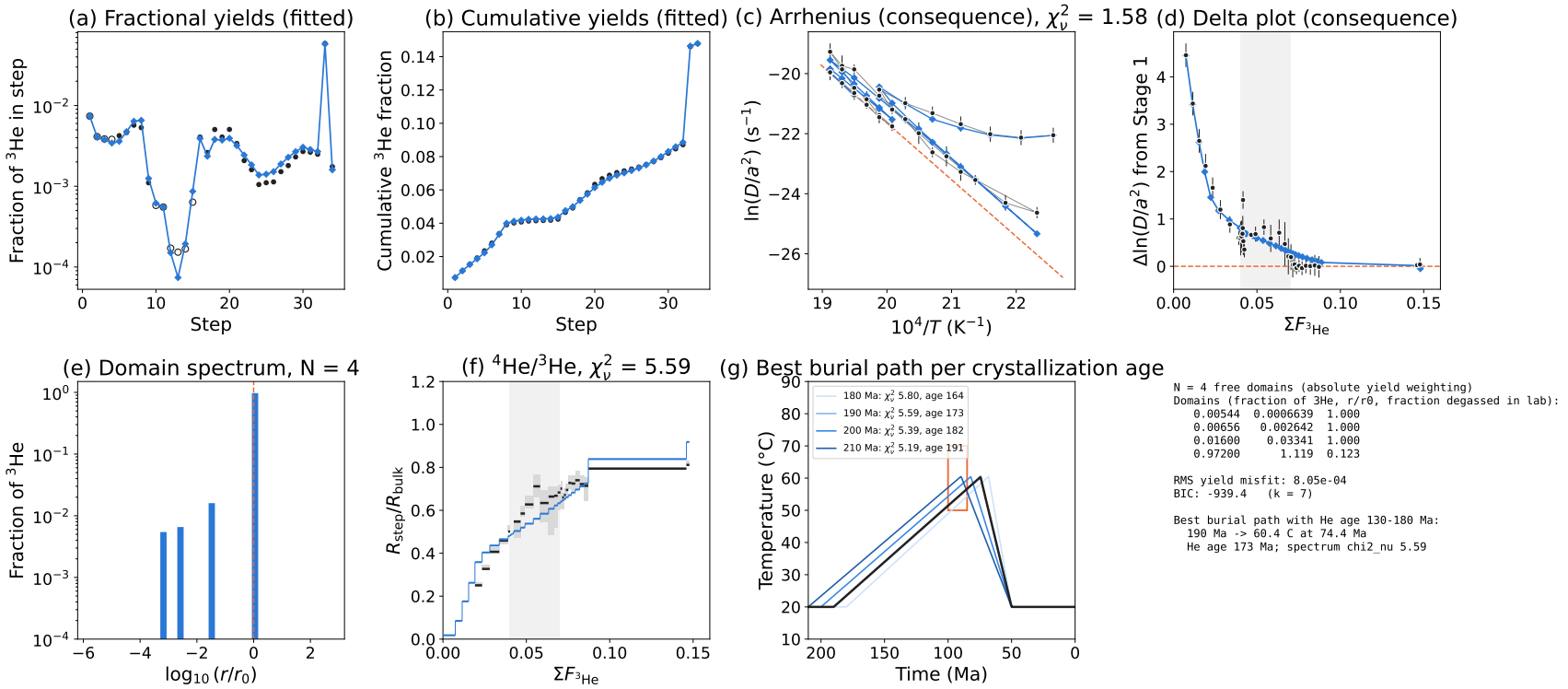


Figure 10: Diagnostic summary of the four-domain model (as Fig. 7).

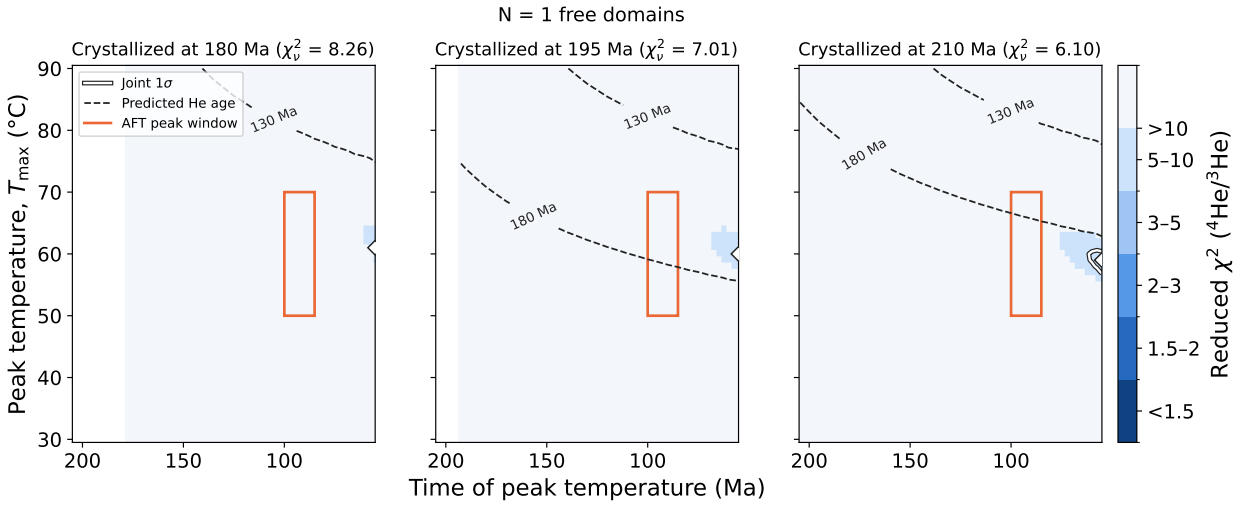


Figure 11: χ^2 maps for the single-domain model (as Fig. 5).

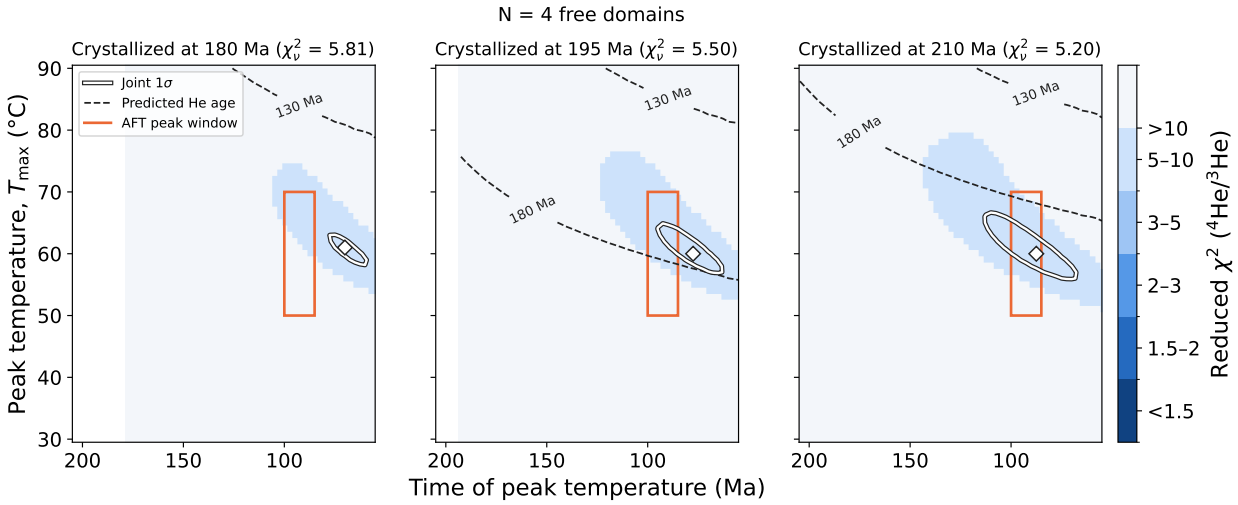


Figure 12: χ^2 maps for the four-domain model (as Fig. 5).